

Structural FEA – Plate Under Load (Stress, Strain & Deformation Study)

Description:

Conducted structural analysis on a flat structural plate under external force to examine total deformation, normal stress, shear stress, and strain distribution. Built the geometry, applied supports and load, and evaluated stress concentration zones and displacement patterns.

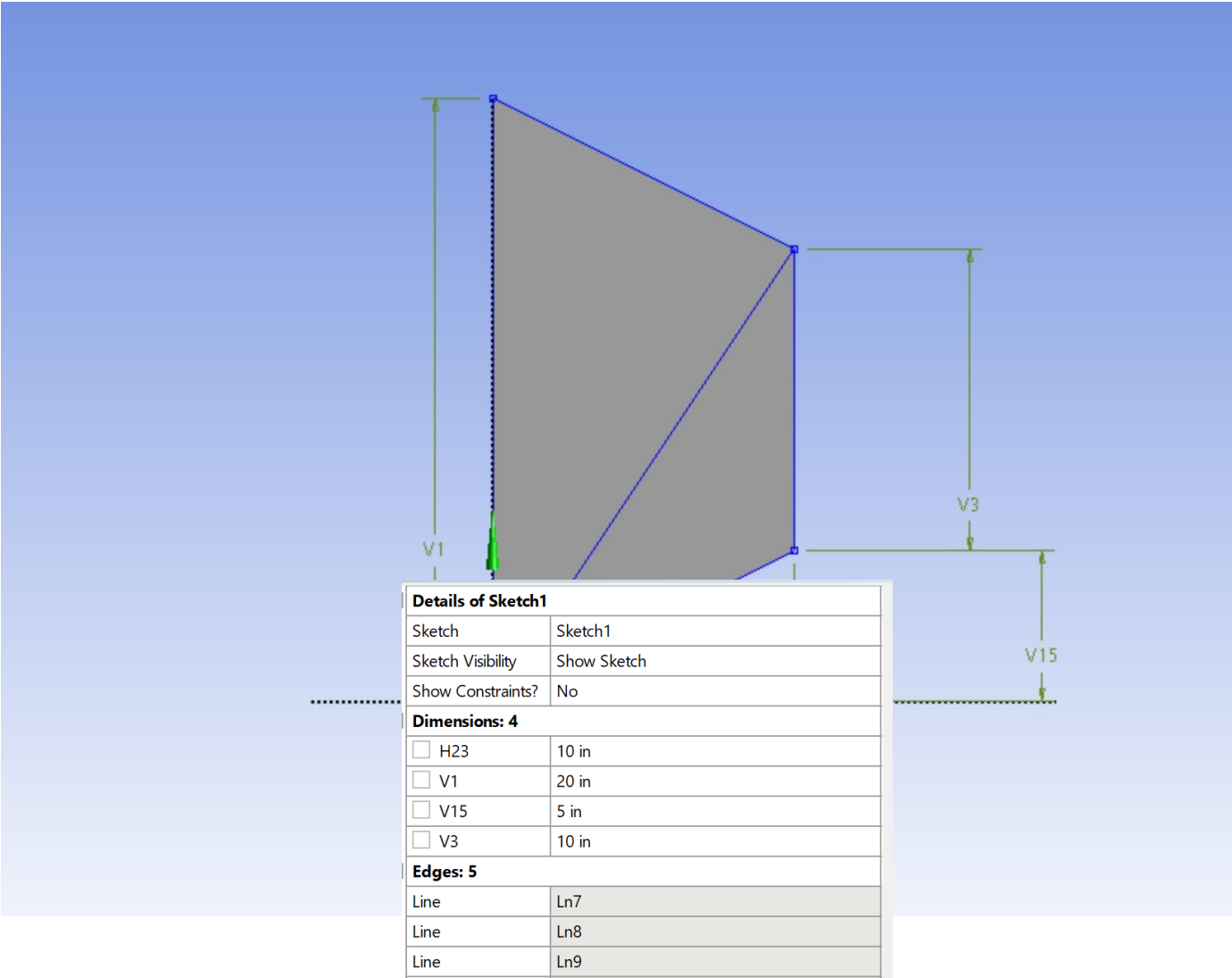
Key Highlights:

- Designed CAD model and refined mesh for accuracy
- Applied fixed supports and directional force loading
- Visualized total deformation, shear stress, normal stress, and strain
- Interpreted material behaviour under load with stress contour mapping
- Demonstrated understanding of structural stiffness and load paths

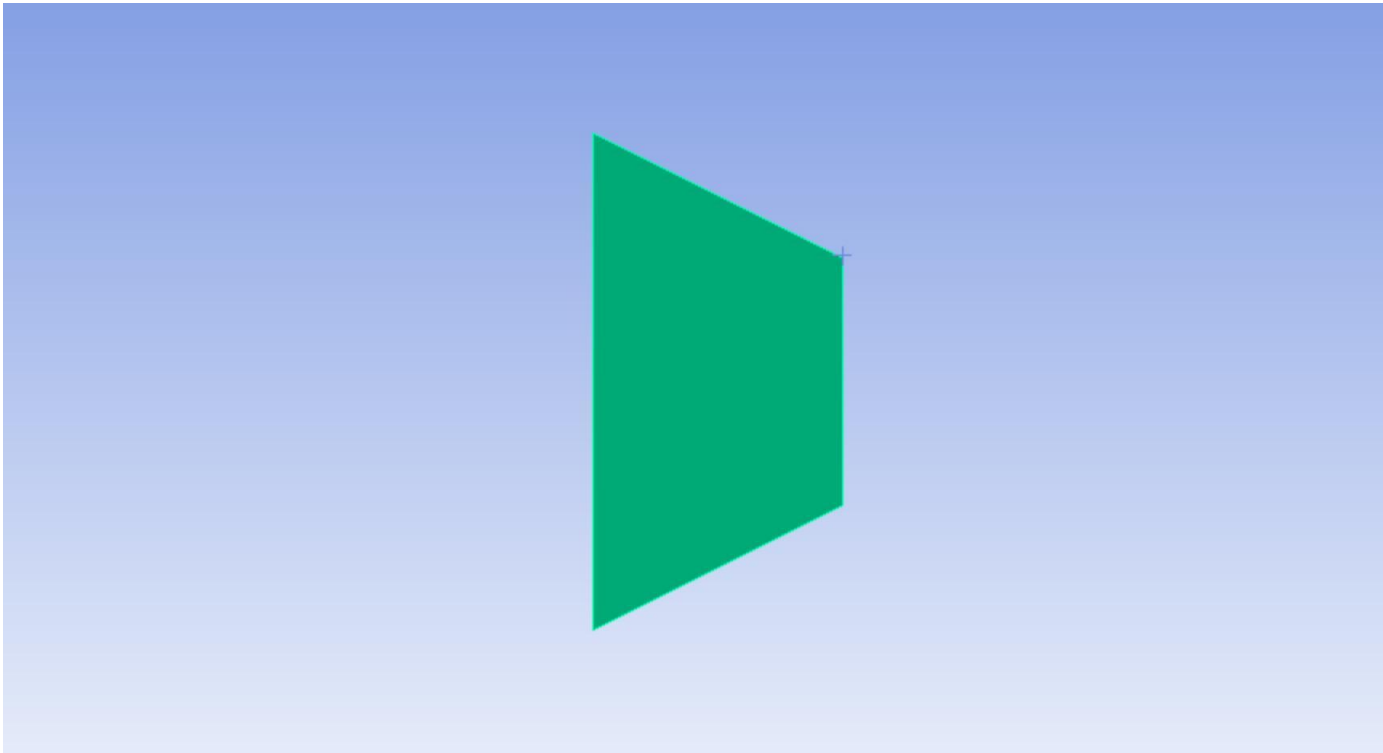
Skills:

Structural Simulation · ANSYS FEA · Stress & Strain Analysis · Mesh Refinement · Engineering Interpretation

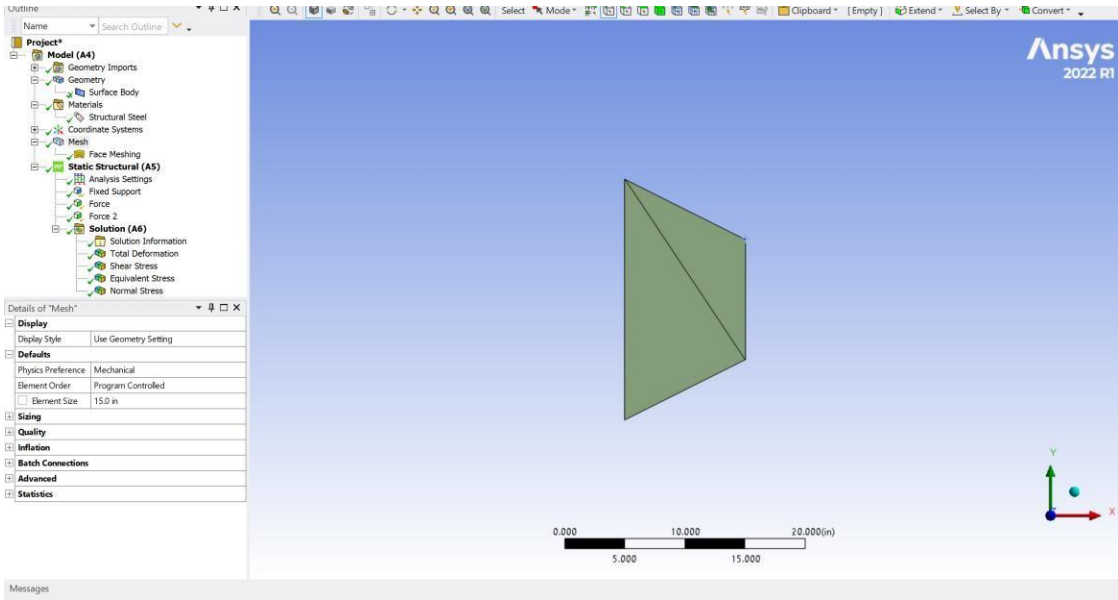
Geometry:



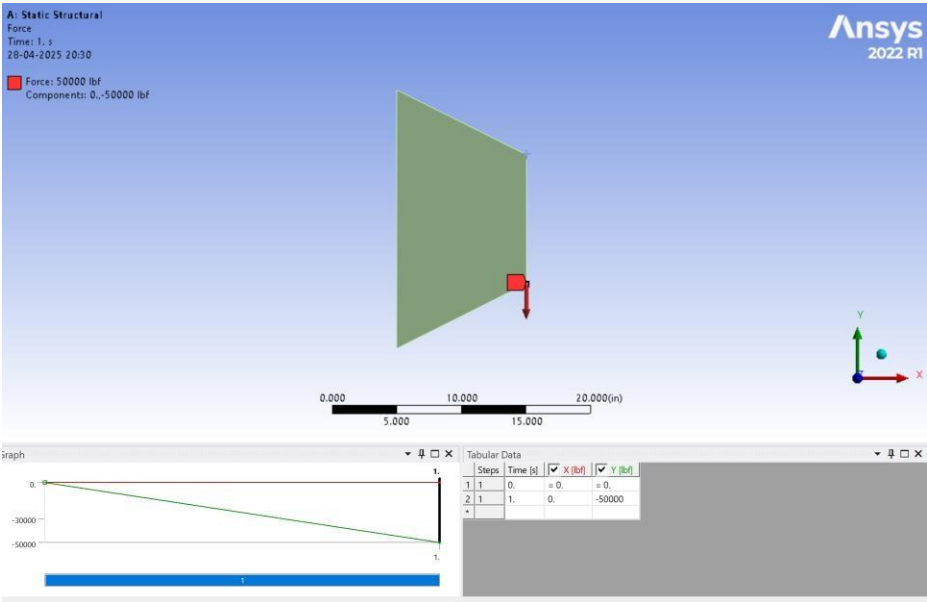
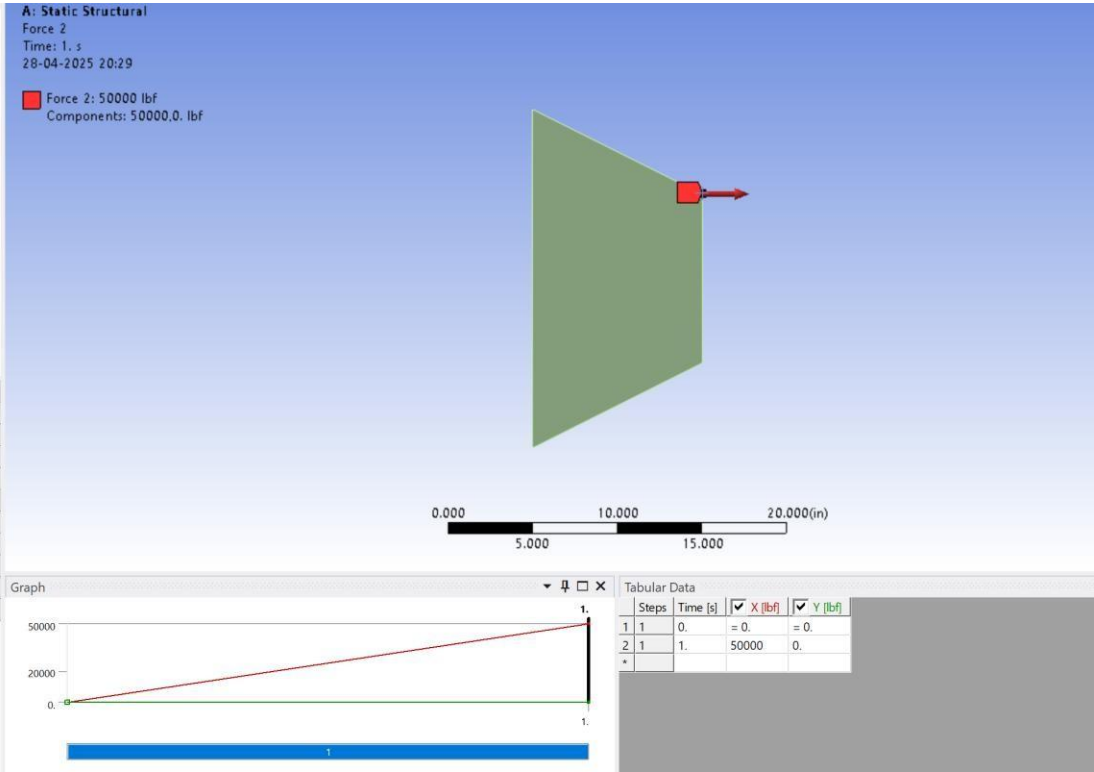
Model:



Mesh:

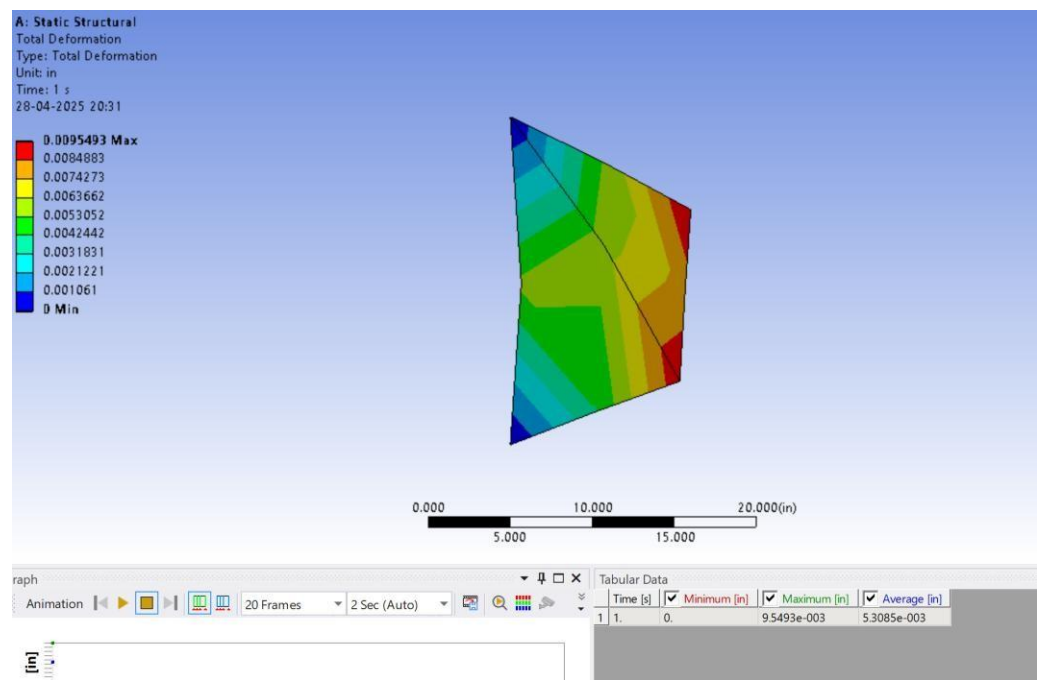


Force:

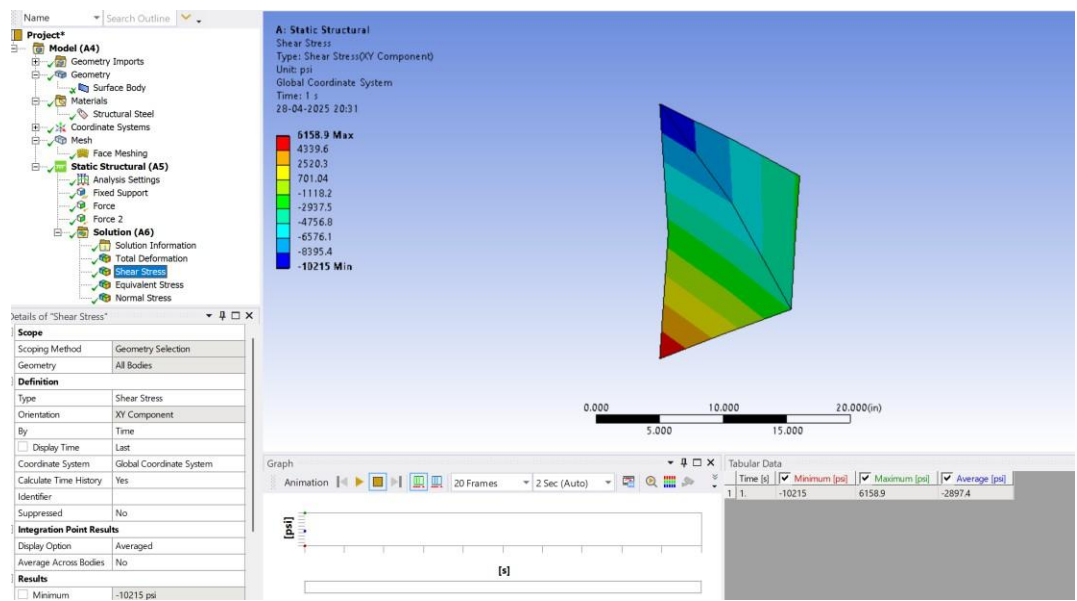


Result:

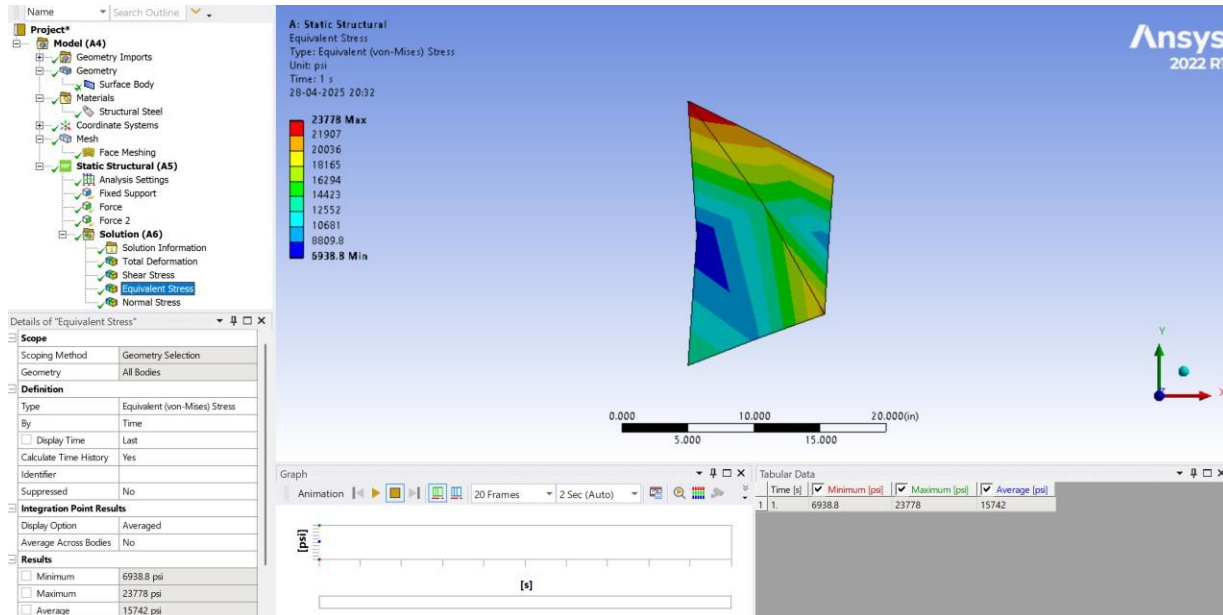
Total Deformation:



Shear stress:



Equivalents stress:



Normal stress:

